

Exam 2

Problem 1

Consider a three-phase voltage source inverter supplying a balanced three-phase RL load. Modify the code provided below to implement a hysteresis current control strategy such that the phase currents track sinusoidal references with an RMS value of 10 A at 60 Hz. The controller shall maintain the current ripple within a maximum band of ± 1 A.

```
%Simulation of a RL load connected to a Controlled-
Voltage-Source
R=5; %Resistance of the RL load
L=0.05; %Inductance of the RL load
io=0; %state variable (load current)
vo=100; %input variable (load voltage)
h=1e-4; %step size
j=0; %variable used to storage the outcomes
t=0; %initial value of the digital time
tmax=1; %maximum simulation time

while t<=tmax, %begining of the simulation loop
    t = t + h; %time incrementation
    io = io + (vo-R*io)*h/L; %state space variable 1
    % storage of the variables
    j = j + 1;
    time(j)=t;
    current(j)=io;
    voltage(j)=vo;
end
figure(1),plot(time,voltage)
figure(2),plot(time,current)
```

Problem 2

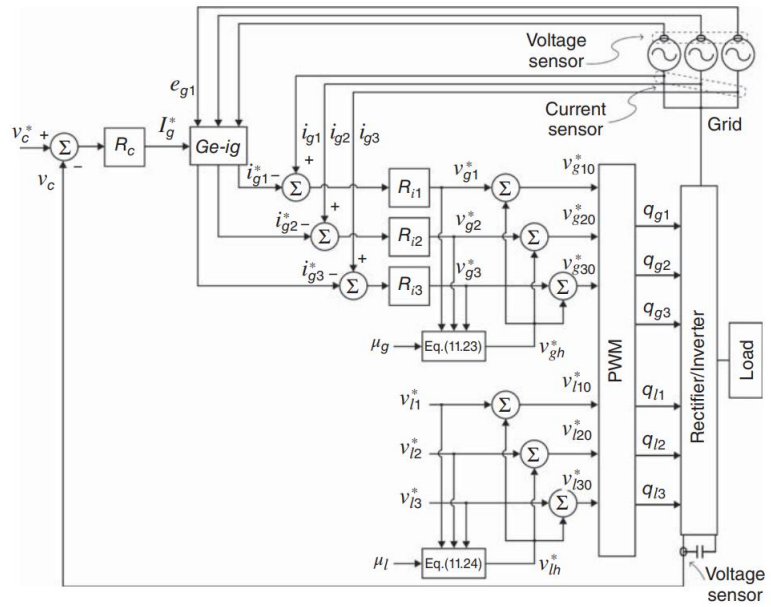
Consider a single-phase AC-DC converter. This configuration is implemented with an H-bridge circuit and a grid reactance interfacing the grid and the controlled rectifier. It is also known that both DC-link voltage and PFC are working as expected. Given the grid voltage: $e_g = 240\sqrt{2}\cos(120\pi t)$, grid reactance of 10 Ohms, and grid current of 24 A_{rms}, determine the amplitude modulation index of this controlled rectifier if the load DC current is 8 A.

Problem 3

A three-phase inverter is employed to supply a two-phase induction machine with orthogonal (90° phase-displaced) voltages. The machine is configured with three terminals: Phase 1 (0° phase angle), Phase 2 (90° phase shift), and a neutral connection. Propose a carrier-based PWM (CPWM) strategy by determining the reference pole voltages v_{10}^* , v_{20}^* , and v_{30}^* for this system. Assume that $v_h^* = -0.5(V_{max}^* + V_{min}^*)$ and derive expressions for V_{max}^* and V_{min}^* .

Problem 4

The control block diagram of a three-phase back-to-back configuration is presented below. Since the regulators R_{i1} , R_{i2} , and R_{i3} are used to control AC variables, they cannot be implemented as simple PI controllers. Modify the system as needed to allow the use of a hysteresis control to control the current in each phase.



Problem 5

- (a) Consider a single-phase back-to-back configuration with a shared leg, meaning that one leg is common to both the input ($120\text{ V}_{\text{rms}}$, 60 Hz) and the output, forming a three-leg topology. If the grid power factor is unity (with $\tilde{I}_g = 10\text{ A}_{\text{rms}}$) and the load requires a frequency equal to that of the grid, if the load power factor is equal 0.5 , determine the load reference voltage that will maximize the current across the shared leg. Assume amplitude modulation index equal to 1 for both input (rectifier) and output (inverter).
- (b) Given the conditions described in (a), determine the minimum DC-link voltage?

